

Supplementary Material: Short-to-Long Functional Connectivity Transfer via Structure-Aware Latent Diffusion

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Abstract. This supplementary document provides additional analyses and implementation details supporting the main paper. We also isolate the effect of longitudinal overlap in ABCD by restricting evaluation to entirely new subjects, excluding follow-up scans of individuals already seen at an earlier session, and confirm the same trends in reconstruction fidelity, cognition prediction, and additional phenotypes. We further show these gains are not tied to a specific evaluation architecture, reproducing the same trends under two different models. We also analyze the shared VAE latent representation, provide qualitative reconstructions, describe the reward and evaluation predictors, and report full architecture, hyperparameter, fine-tuning, and compute details for reproducibility.

1 Phenotype Prediction Models

This section describes the predictive models used within the SALD framework for reward-guided training and evaluation. During diffusion fine-tuning, a differentiable phenotype predictor provides the reward signal used to guide generation toward behaviorally informative connectivity patterns. To avoid circular evaluation, phenotype prediction performance is assessed using a separate model trained independently from the reward predictor on the same training and validation split as SALD, with the test set held out throughout. The following subsections detail the kernel ridge regression model used for reward computation and the linear ridge regression model used for evaluation.

1.1 Kernel Ridge Regression (Reward Predictor)

During SALD fine-tuning, we require a differentiable model that maps a generated functional connectivity (FC) matrix to the predicted phenotype. This model provides the reward signal used to guide the diffusion model toward generating phenotype-informative connectivity patterns. For this purpose we employ a Kernel Ridge Regression (KRR) predictor trained on long-scan connectivity matrices FC_{long} .

Kernel methods have been widely used for phenotype prediction from functional connectivity and have been shown to perform comparably to deep neural networks in this setting [2]. We use the correlation kernel, which measures similarity between connectivity patterns while being invariant to global shifts and scaling that may arise from scanner effects or generative model artifacts. For vectorized upper-triangle connectivity vectors $x_i \in \mathbb{R}^{4950}$ the kernel is

$$k(x_i, x_j) = \frac{(x_i - \bar{x}_i \mathbf{1})^\top (x_j - \bar{x}_j \mathbf{1})}{\|x_i - \bar{x}_i \mathbf{1}\| \|x_j - \bar{x}_j \mathbf{1}\|}. \quad (1)$$

In the classical dual formulation, KRR predicts the phenotype as

$$\hat{y}(x) = \eta_0 + \sum_{i=1}^n \eta_i k(x_i, x), \quad (2)$$

where $\{x_i\}_{i=1}^n$ are the training connectivity vectors. While this representation is correct, evaluating the kernel against all training samples during diffusion fine-tuning would be computationally inefficient.

Instead, we use the equivalent primal representation. Specifically, the correlation kernel can be expressed through the normalized feature map

$$\phi(x) = \frac{x - \bar{x} \mathbf{1}}{\|x - \bar{x} \mathbf{1}\|_2}, \quad (3)$$

which yields the linear predictor

$$\hat{y}(x) = \alpha_0 + \boldsymbol{\alpha}^\top \phi(x). \quad (4)$$

The parameters $(\alpha_0, \boldsymbol{\alpha})$ are obtained by solving ridge regression with ℓ_2 penalty $\rho \|\boldsymbol{\alpha}\|_2^2$, where the regularization parameter ρ is selected using the validation set.

This representation is particularly convenient for SALD fine-tuning. The correlation kernel mitigates trivial reward reward hacking because it is invariant to global shifts and scaling of the connectivity vector. Consequently, the predictor does not reward uniformly increasing FC magnitudes but instead depends on the relative connectivity structure, encouraging the model to remain close to the empirical manifold of functional connectivity patterns. During diffusion training, the predictor parameters $(\alpha_0, \boldsymbol{\alpha})$ remain fixed while the feature map $\phi(\cdot)$ is differentiable with respect to the generated FC. This allows gradients of the phenotype prediction error to propagate through the connectivity representation, providing a stable reward signal that guides the DiT backbone toward generating phenotype-informative connectivity patterns.

1.2 Linear Ridge Regression (Evaluation Predictor)

To assess whether the signal captured by SALD generalizes beyond the specific reward model, rather than reflecting reward-model exploitation, we evaluate phenotype prediction using a separate linear ridge regression model. This predictor

is trained independently from the reward model using the same training data but a different functional form, reducing the risk of circular evaluation.

Specifically, we fit a linear ridge regression model using the flattened long-scan connectivity vectors x as input,

$$\min_{b, \mathbf{w}} \|y - b - \mathbf{w}^\top x\|_2^2 + \rho' \|\mathbf{w}\|_2^2, \quad (5)$$

where b is the intercept and $\mathbf{w}$ denotes the regression coefficients. The regularization parameter ρ' is selected using the validation set.

Using a model with a different inductive bias from the kernel-based reward predictor provides an additional safeguard against reward overfitting, ensuring that improvements in phenotype prediction reflect meaningful structure in the generated functional connectivity rather than artifacts specific to the reward model.

2 Additional Experiments

This section presents supplementary analyses that further validate the SALD framework. Most importantly, we deliver on the claim made in the main paper that SALD’s gains extend to entirely new subjects: when evaluation is restricted to individuals unseen during training, the same trends hold across reconstruction fidelity, cognition prediction, and additional phenotypes. We further examine the shared latent representation learned by the VAE, test robustness under alternative downstream predictors, and provide additional qualitative examples of generated functional connectivity.

2.1 Additional Generalization Analysis

Because the ABCD study is longitudinal, a subject scanned at baseline may also return for a follow-up session roughly two years later. Under the scan-level partition used in the main paper, these two independent acquisitions from the same individual can fall in different splits, so a baseline session may appear in training while the corresponding follow-up session appears in testing. To verify that the improvements reported for SALD are not driven by this subject-level overlap, we re-evaluate the *same trained models* on a restricted test set that excludes every held-out scan whose subject also contributes scans to training. No model is retrained; only the evaluation set changes. This leaves 338 scans from subjects entirely unseen during training, and all other settings, including the 20-minute reference target, the conditioning inputs, and the cross-architecture Linear Ridge Regression evaluation predictor, match the main paper.

Across reconstruction fidelity, cognition prediction, and generalization to additional phenotypes, SALD reproduces the same qualitative trends in this more stringent setting, demonstrating that its gains extend to subjects the model has never encountered.

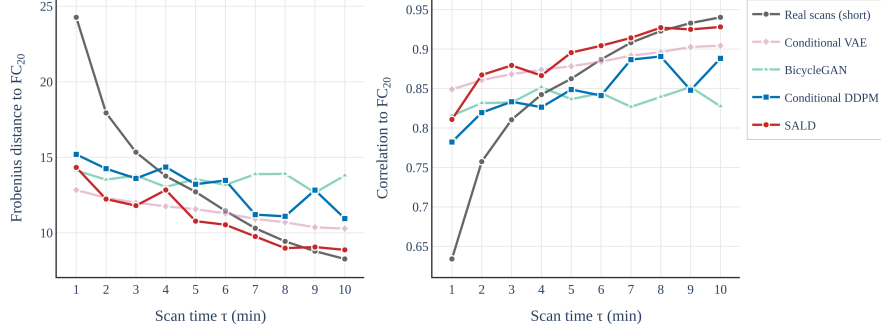


Fig. 1: Reconstruction fidelity to the 20-minute reference FC (FC_{20}) on entirely new subjects (338 held-out scans, after removing test scans whose subjects also appear in training), across input scan durations τ . Left: Frobenius distance ($\downarrow$); right: Pearson correlation ($\uparrow$). As in the main paper, SALD achieves the strongest overall agreement with FC_{20} among the generative baselines and consistently improves over Conditional DDPM, while Conditional VAE remains competitive in Frobenius distance. Real short-scan FCs are included as a non-generative reference.

Reconstruction fidelity. Figure 1 reports Frobenius distance and correlation to the 20-minute reference FC (FC_{20}). Consistent with the main results, SALD achieves the strongest overall agreement with FC_{20} among the generative baselines and consistently improves over Conditional DDPM, with the largest margins at short, noisy scan durations. Conditional VAE remains competitive in Frobenius distance, in line with its explicit reconstruction objective, while the real short-scan FCs (a non-generative reference) only approach the generative methods at longer durations.

Cognition prediction. Figure 2 evaluates prediction of the NIH Toolbox Total Cognition Composite Score, the phenotype SALD was fine-tuned on, using MSE, correlation, and recovered predictive variance. SALD again provides the strongest overall performance among the generative baselines, improving over the observed short-scan FCs and recovering more of the long-scan predictive variance than Conditional DDPM. Its advantage is most pronounced at the shortest durations, where the observed FC is least reliable; once roughly six minutes of data are available, SALD ties the real short-scan FC. This contrasts with the main setting, where prior sessions from the same subject let SALD improve on the short scan across durations: here, with no prior information about a new subject to draw on, the short scan already carries enough signal to match it once enough data accumulate. The benefit of SALD therefore concentrates exactly where it is most useful, recovering reliable connectivity when scan time is scarce. As in the main setting, this steering signal is weak: long-scan FC explains only a small fraction of phenotype variance, yet SALD consistently concentrates its improvements on this explainable structure. BicycleGAN shows unstable phe-

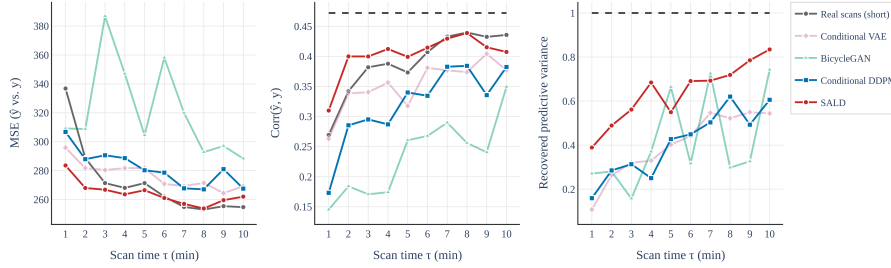


Fig. 2: Cognition prediction (NIH Toolbox Total Cognition Composite Score) on entirely new subjects across scan durations τ : MSE ($\downarrow$), correlation ($\uparrow$), and recovered predictive variance ($\uparrow$). SALD provides the strongest overall performance among the generative baselines, with the clearest gains in the short-scan regime and more recovered long-scan predictive variance than Conditional DDPM; BicycleGAN shows unstable phenotype utility despite stable reconstruction error. Dashed lines denote the 20-minute reference.

notype utility across durations despite comparatively stable reconstruction error.

Generalization to additional phenotypes. Following the main paper, we further evaluate UPPS negative urgency, Child Behavior Checklist (CBCL) Total Problems, and Sleep Disturbance Scale for Children (SDS) total sleep disturbance, none of which were used during fine-tuning. Fig. 3 shows that, on entirely new subjects, SALD generally recovers more of the 20-minute predictive variance and achieves comparable or stronger phenotype correlations than Conditional DDPM, confirming that its phenotype utility extends beyond both the fine-tuning target and the subjects seen during training.

Together, these results show that the fidelity and phenotype-utility advantages of SALD persist when test subjects are entirely held out, supporting the conclusion that the method recovers genuinely transferable, phenotype-relevant connectivity structure rather than exploiting subject overlap between training and evaluation.

2.2 Analysis of the Shared Latent Representation

Our framework assumes that functional connectivity (FC) matrices computed from scans of two different durations can be embedded into a shared latent representation using a single VAE encoder. This design encourages the encoder to learn a representation that captures subject-specific connectivity structure while being invariant to scan duration.

To achieve this, the VAE is trained jointly on paired short- and long-duration FC matrices. Given a pair $(FC_\tau, FC_{\text{long}})$ for the same subject, both inputs are encoded by the same encoder network and reconstructed through a shared

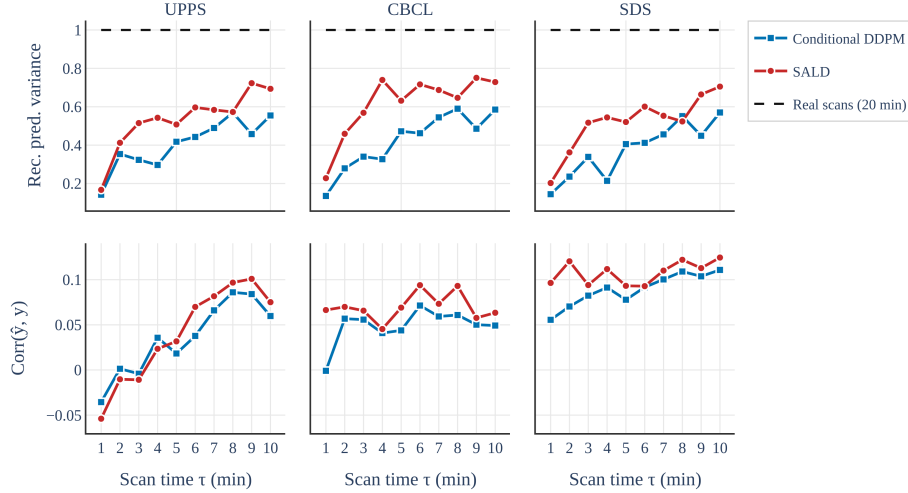


Fig. 3: Generalization to additional phenotypes on entirely new subjects. Recovered predictive variance (top) and predicted–observed correlation (bottom) for UPPS negative urgency, CBCL Total Problems, and SDS total sleep disturbance, none used during fine-tuning. SALD generally recovers more 20-minute predictive variance and achieves comparable or stronger correlations than Conditional DDPM across all three phenotypes.

decoder. The reconstruction loss is applied to both inputs, and the resulting objective encourages the encoder to represent both noisy short scans and more stable long scans within a common latent manifold.

A potential concern is that the encoder might instead learn separate latent regions for short and long scans due to their different noise characteristics. To evaluate whether this occurs in practice, we analyze the structure of the learned latent space.

Latent Space Alignment. We project latent representations of test subjects using t-SNE applied to the VAE latent vectors. For each subject we compute two embeddings: one corresponding to the latent encoding of the long scan and another corresponding to the latent encoding of the short scan. Figure 4 visualizes these embeddings for three different short-scan durations.

Across all durations we observe strong overlap between the latent representations of short and long scans for the same subject. Points corresponding to the two embeddings are nearly co-located, indicating that the encoder maps both observations to highly similar latent vectors. This behavior suggests that the encoder successfully captures subject-specific connectivity structure while suppressing scan-duration noise.

Importantly, this property is critical for the subsequent diffusion model. Because both scan durations occupy a common latent manifold, the diffusion model

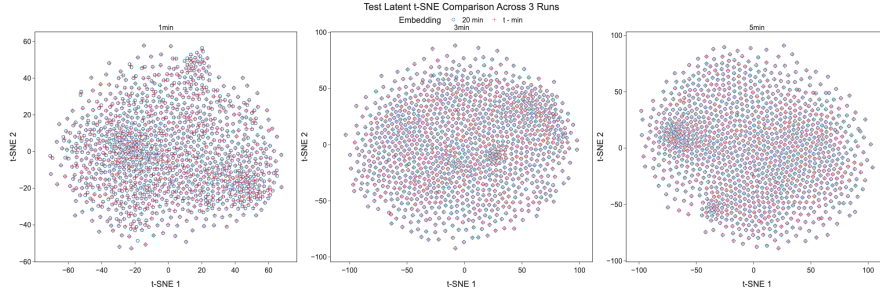


Fig. 4: t-SNE visualization of VAE latent representations for test subjects. Blue circles correspond to latent embeddings of long scans, while red markers correspond to embeddings of short scans for the same subjects. Across all short-scan durations (1, 3, and 5 minutes), the two embeddings are nearly co-located for each subject, indicating that the encoder maps both observations to a shared latent manifold. This empirical evidence supports the assumption that the VAE learns a duration-invariant representation of functional connectivity.

can operate as a conditional refinement process that transforms the latent representation of a short scan into the latent representation of a long scan while preserving subject identity.

2.3 Alternative Downstream Predictors

In the main paper, phenotype prediction performance is evaluated using a linear ridge regression model trained on the 20-minute reference functional connectivity (FC) matrices, while the reward signal used during diffusion fine-tuning is produced by a kernel ridge regression (KRR) predictor. To provide additional context, we first report results obtained using the same correlation-kernel KRR predictor used to guide the steering process. These results represent the evaluation setting most closely aligned with the reward signal used during steering.

To further verify that the improvements observed for SALD are not an artifact of the linear/kernel-based predictor used for phenotype scoring, we also evaluate phenotype prediction using a gradient-boosted tree model. The model is trained on the vectorized upper-triangle features of the FC matrices using a fixed configuration: `n_estimators=200`, `learning_rate=0.05`, `max_depth=3`, `subsample=0.8`, and `colsample_bytree=0.5`. While tree-based models are generally not optimal for high-dimensional functional connectivity prediction, they provide a useful robustness check because they represent a substantially different model class from the linear (LRR) and kernel-based (KRR) predictors used in the main experiments.

Figures 5 and 6 summarize the results across scan durations. Consistent with the main results, SALD improves phenotype prediction performance over the Conditional DDPM baseline across all metrics (MSE, correlation, and R^2) and scan durations. The same qualitative trends hold even under the tree model

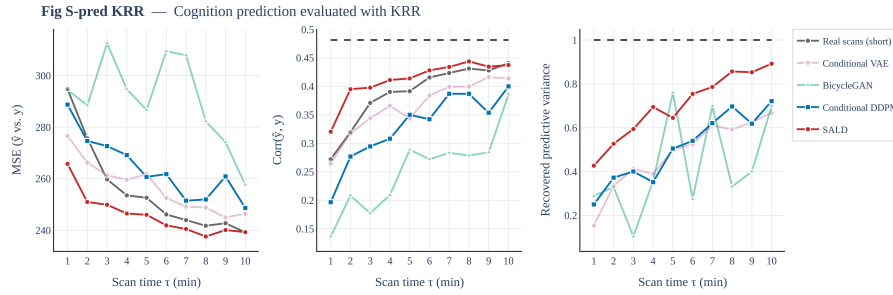


Fig. 5: Phenotype prediction performance when evaluated using the correlation-kernel ridge regression (KRR) predictor employed during reward-guided diffusion training. Results are reported across scan durations using three metrics: mean squared error (MSE), correlation, and R^2 . SALD consistently improves predictive performance over the Conditional DDPM baseline and narrows the gap with the long-scan reference (dashed line), demonstrating that the steering mechanism successfully preserves phenotype-discriminative connectivity structure.

predictor, demonstrating that the improvements produced by SALD do not depend on the specific architecture of the evaluation model but instead reflect the recovery of phenotype-relevant structure in the generated functional connectivity.

2.4 Qualitative Connectivity Examples

We provide qualitative examples of generated functional connectivity (FC) matrices to illustrate the behavior of SALD relative to the Conditional DDPM baseline. Fig. 7 shows representative reconstructions for two subjects. The short 5-minute scans introduce noise that partially obscures the subject-specific structure visible in the 20-minute reference FC. Both generative approaches recover plausible connectivity structure, and the differences between Conditional DDPM and SALD are visually subtle. Nevertheless, these small structural variations significantly affect downstream phenotype prediction. This illustrates an important property of functional connectivity: relatively small changes in correlation structure can carry meaningful behavioral information, motivating generation methods that preserve predictive signal rather than only distributional similarity.

To show that this behavior is consistent across subjects, Fig. 8 presents reconstructions for several additional subjects using the same column layout. Across subjects, the short-scan FC matrices exhibit noticeable noise and attenuation of subject-specific connectivity patterns, while SALD produces reconstructions that more closely resemble the long-scan reference than the Conditional DDPM baseline.

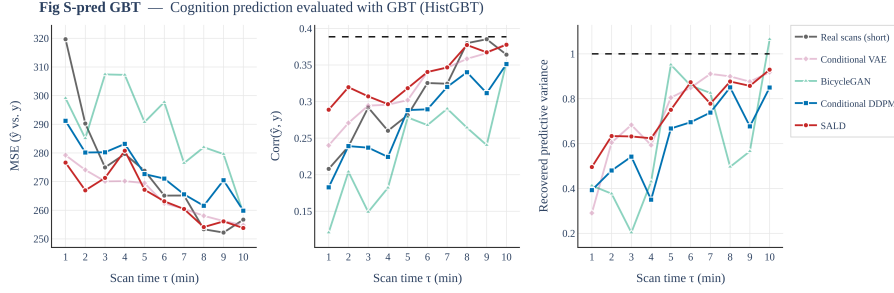


Fig. 6: Phenotype prediction evaluated using an tree model regressor trained on vectorized functional connectivity features. Although tree-based models are typically less effective for high-dimensional FC prediction, SALD still improves prediction performance over Conditional DDPM across scan durations and evaluation metrics. This result confirms that the gains observed for SALD are not specific to the linear or kernel-based predictors used in the main experiments.

3 SALD Architecture Details

This section provides the architectural details of the models used in the SALD framework. In particular, we describe the variational autoencoder (VAE) used to encode functional connectivity matrices into a compact latent representation and the diffusion transformer backbone that operates in this latent space.

3.1 Variational Autoencoder

To obtain a compact latent representation of functional connectivity (FC) matrices, we train a variational autoencoder (VAE) [4]. Each FC matrix is vectorized to shape $(1, 4950)$ and encoded into a latent tensor of shape $(4, 618)$. The same encoder is used for both short- and long-scan inputs, ensuring a shared latent space for diffusion training. The encoder-decoder follows a U-Net architecture [6]; full architectural details are provided in Tab. 1.

We assume a Gaussian latent prior $p(z) = \mathcal{N}(0, I)$ and approximate posterior $q_\phi(z | FC) = \mathcal{N}(\mu_\phi(FC), \sigma_\phi(FC))$. The VAE is trained using the standard reconstruction-plus-KL objective

$$\mathcal{L}_{\text{VAE}} = \frac{1}{2} \sum_{t \in \{\text{short}, \text{long}\}} \left[\|FC_t - \widehat{FC}_t\|_2^2 + \beta_{\text{KL}} D_{\text{KL}}(q_\phi(z | FC_t) \| p(z)) \right], \quad (6)$$

with $\beta_{\text{KL}} = 10^{-7}$. During diffusion training we use the encoder posterior mean $\mu_\phi(FC)$ as the latent representation.

3.2 Diffusion Transformer Backbone

The diffusion model operates in the VAE latent space using a 1D Diffusion Transformer (DiT) [5]. Architectural hyperparameters of the transformer are

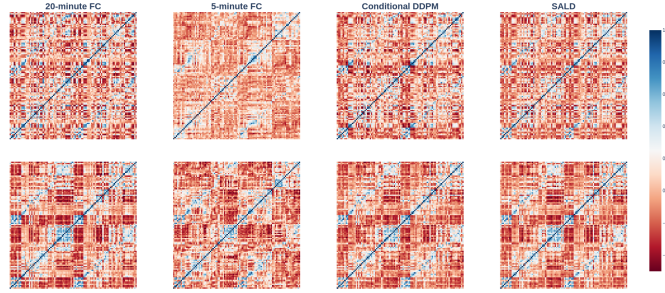


Fig. 7: Qualitative connectivity reconstructions for two subjects. From left to right: 20-minute reference FC, noisy 5-minute FC, Conditional DDPM reconstruction, and SALD. Differences between the generative methods are visually subtle, but these small structural variations are sufficient to influence downstream phenotype prediction, highlighting the importance of preserving behaviorally relevant connectivity patterns.

Table 1: VAE architecture used to encode vectorized functional connectivity matrices from $(1, 4950)$ to a latent representation of shape $(4, 618)$. The encoder follows a U-Net-style design with progressive channel expansion and strided convolutions, mirrored by transpose convolutions in the decoder.

Component	Value
Input channels	1
Input representation shape	$(1, 4950)$
Latent channels	4
Encoder down channel schedule	$[128, 256, 512, 512]$
Bottleneck channel schedule	$[512, 512]$
Number of heads	8
Normalization type	Group Norm
Normalization Groups	32
Downsampling conv kernel sizes	$[4, 4, 4]$
Downsampling conv strides	$[2, 2, 2]$
Upsampling transpose kernel sizes	$[4, 4, 4]$
Upsampling transpose strides	$[2, 2, 2]$
Transpose output padding	$[0, 1, 1]$

summarized in Tab. 2, while details of the conditioning modules are provided in Tab. 3.

The latent representation of the short scan is concatenated with the noisy latent representation of the long scan along the channel dimension, forming a sequence of length 618 with 8 channels. We use a patch size of 1, allowing the transformer to operate directly at the latent resolution.

Conditioning information is incorporated through adaptive layer normalization (AdaLN). Structural connectivity (SC) is first embedded using a two-layer MLP to produce a 128-dimensional representation. Subject covariates are en-

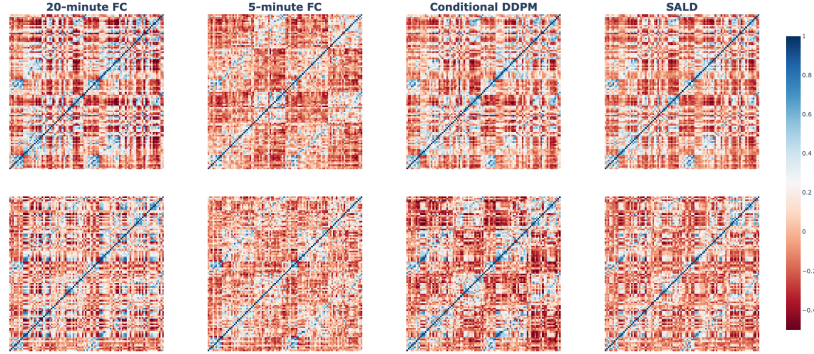


Fig. 8: Further qualitative examples of functional connectivity reconstruction. Each row corresponds to a different subject. From left to right: the 20-minute reference FC, the noisy 5-minute short-scan FC, the reconstruction produced by the Conditional DDPM baseline, and the reconstruction generated by SALD. Consistent with Fig. 7, SALD more faithfully restores the structure present in the long-scan reference, and these subtle differences in connectivity structure carry phenotype-relevant information.

Table 2: 1D Diffusion Transformer architecture used in the VAE latent space. The model processes sequences of length 618 formed by concatenating the short-scan latent representation with the noisy long-scan latent representation along the channel dimension.

Component	Value
Input sequence length	618
Input sequence channels (after concat)	8
Output sequence channels	4
Patch size	1
Patch embedding layer type	Linear
Positional embedding type	Sinusoidal
Number of transformer layers	12
Hidden size	512
Attention heads	8
Head dimension	64

coded by combining a 126-dimensional sinusoidal embedding of age with a one-hot encoding of sex, yielding a 128-dimensional vector that is projected through a two-layer MLP.

The SC embedding, covariate embedding, and diffusion timestep embedding are fused to produce the modulation parameters used by AdaLN to condition the transformer blocks. This mechanism allows the model to integrate structural and demographic information during the denoising process.

Table 3: Embedding and projection modules for conditioning signals. Structural connectivity, subject covariates (age and sex), and diffusion timestep embeddings are projected through MLPs and fused to produce AdaLN modulation parameters for the transformer blocks.

Component	Value
SC embedding dimension	128
Timestep embedding dimension	512
Timestep projection MLP	2-layer MLP (SiLU)
Class/covariate projection MLP	2-layer MLP (SiLU)
Context projection MLP (FiLM context)	2-layer MLP (SiLU)

4 Training Details

Each short-scan duration τ was trained independently using the same training/validation/test partition. The reward predictor remained fixed across all short-scan settings. For each τ , training proceeded sequentially: first the VAE was trained, followed by the diffusion model, and finally reward-aligned fine-tuning. A summary of the training hyperparameters is provided in Tab. 4.

4.1 Training Hyperparameters

Because multiple models were trained across several short-scan durations, we employed early stopping to avoid manual intervention during training. Both the VAE and diffusion models were trained with validation-based early stopping, terminating training after a fixed number of consecutive epochs without improvement and selecting the checkpoint with the best validation performance.

In contrast, the fine-tuning stage was run for a fixed budget of 9 epochs, with validation performed every 0.2 epochs. The final model was selected based on the minimum validation loss.

To improve computational efficiency and reduce memory usage, all models were trained using mixed-precision training with **bf16**.

4.2 Reward Steering Configuration

The most computationally demanding and sensitive stage of the pipeline is the reward-aligned adaptation performed using DRaFT [1]. To efficiently steer the pretrained diffusion model, we apply Low-Rank Adaptation (LoRA) [3] to the linear layers of the transformer backbone.

Specifically, each adapted weight matrix is parameterized as

$$\mathbf{W}_{\text{adapted}} = \mathbf{W}_0 + \frac{\alpha}{r} \mathbf{A} \mathbf{B}, \quad (7)$$

where $\mathbf{W}_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ is the frozen pretrained weight, $\mathbf{A} \in \mathbb{R}^{d_{\text{out}} \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d_{\text{in}}}$ are trainable low-rank factors, r is the LoRA rank, and α is a scaling coefficient controlling the magnitude of the adaptation.

Table 4: Training hyperparameter template.

Stage	Hyperparameter	Value
[VAE]	Optimizer	Adam
[VAE]	Learning rate	$2e - 4$
[VAE]	Batch size	64
[VAE]	Max. Epochs	200
[VAE]	VAE KL weighting	$1e - 7$
[VAE]	Early Stopping	15
[VAE]	LR scheduler type	ReduceLROnPlateau
[VAE]	Gradient clipping Max Norm	1
[VAE]	Mixed precision mode	bf16
[Diffusion]	Optimizer	Adam
[Diffusion]	Learning rate	$8e - 5$
[Diffusion]	Batch size	64
[Diffusion]	Max Epochs	200
[Diffusion]	Diffusion steps	1000
[Diffusion]	Noise schedule configuration	Linear
[Diffusion]	Early Stopping patience	25
[Diffusion]	LR scheduler type	Constant
[Diffusion]	Mixed precision mode	bf16
[Finetune]	Optimizer	AdamW
[Finetune]	Learning rate	$4e - 4$
[Finetune]	Weight decay	0
[Finetune]	batch size	64
[Finetune]	Epochs	9
[Finetune]	Warmup scheduler type	Linear
[Finetune]	Warmup duration	125 steps
[Finetune]	Gradient clipping Max Norm	1
[Finetune]	Mixed precision	bf16
[Finetune]	Logging/validation interval	0.2 epochs

LoRA adapters are applied to the attention query, key, and value projections, the attention output projection, the MLP projections, and the final output projection layer. Empirically, we found the final output projection to have the largest influence on phenotype-aligned generation, and therefore allocate a larger LoRA capacity to this layer.

Following [1], we adopt a variant of DRaFT using partial backpropagation through the final $K = 5$ denoising steps. To stabilize the reward signal during generation, we additionally incorporate the low-variance (LV) sampling strategy with four repeated evaluations per sample. Diffusion sampling is performed using 50 DDIM denoising steps.

Unlike settings where reward hacking can be controlled through weight decay or limited adaptation capacity, we found that in functional connectivity synthesis the generator can still exploit the reward predictor without appropriate regu-

Table 5: Hyperparameters used for reward-aligned LoRA fine-tuning with DRaFT. The table reports the LoRA rank r and scaling factor α applied to different transformer projections, together with the steering loss weights (distillation λ and anchoring γ), diffusion sampling configuration, and low-variance evaluation settings used during fine-tuning.

Component	Value
LoRA (attention QKV projections)	$r = 8, \alpha = 2$
LoRA (attention output projection)	$r = 8, \alpha = 2$
LoRA (final output projection)	$r = 32, \alpha = 16$
LoRA (MLP projections)	$r = 8, \alpha = 2$
Distillation weight λ	1.2
Anchoring-regularization weight γ	1
DDIM Steps	50
Partial denoising depth (K)	5
Low-variance repeats	4

larization. To mitigate this effect, we include the noise anchoring regularization term. The full set of fine-tuning hyperparameters is summarized in Sec. 4.2.

5 Generative Baseline Details

For transparency and reproducibility, we report the configuration of the two generative baselines compared against SALD: a Conditional VAE and a BicycleGAN. Both generate the 20-minute reference FC matrix (vectorized upper triangle, 4950 dimensions) conditioned on the short-scan FC, structural connectivity (SC), and demographic covariates. As with SALD, each baseline is trained independently per short-scan duration on the same data partition, FC matrices are used in raw (non-standardized) form.

Conditional VAE. The Conditional VAE follows a 1D U-Net [6] encoder-decoder with residual blocks and self-attention, trained with the standard β -VAE objective [7] ($\beta = 2 \times 10^{-3}$). The encoder takes the channel-wise concatenation of the target FC, short-scan FC, and SC vector; the decoder reconstructs the 20-minute FC from z with the conditioning added at the latent level. Covariates are embedded via a two-layer MLP and injected into the residual blocks. Details are given in Tab. 6.

BicycleGAN. The BicycleGAN baseline [8] couples a conditional VAE-GAN path with a conditional latent-regressor path. All modules are fully connected MLPs operating on flattened vectors, with a 10028-dim condition (short-scan FC + SC + covariates) and latent dimension 256. The generator loss combines adversarial, L1 reconstruction, KL, and latent-regression terms with weights 1.0, 7.5, 0.01, and 1.5, respectively. Details are given in Tab. 7, and training hyperparameters for both baselines in Tab. 8.

Table 6: Conditional VAE architecture (1D U-Net encoder-decoder).

Component	Value
Input channels (after concat)	3 (target FC + short FC + SC)
Encoder down channel schedule	[128, 256, 512, 512]
Bottleneck channel schedule	[512, 512]
Latent channels	4
Down-sampling stages	3 (stride-2 Conv1d, kernel 4)
Residual blocks per stage	2
Self-attention	last down-stage only
Attention heads	8
Normalization	Group Norm (32 groups)
Covariate embedding	10-dim $\rightarrow$ 512-dim, 2-layer MLP (SiLU)
Latent conditioning	SC/short-scan projected, added to z
Activation	SiLU

Table 7: BicycleGAN architecture. c is the 10028-dim condition, y the 4950-dim FC, and $z \in \mathbb{R}^{256}$.

Module	Input dim	Hidden dims	Output dim	Activation
Generator $G(c, z)$	10028 + 256	1024, 1024, 512	4950	ReLU
Encoder $E(c, y)$	10028 + 4950	1024, 512	256 ($\mu, \log \sigma^2$)	ReLU
Discriminator $D(c, y)$	10028 + 4950	1024, 512	1 (logit)	LeakyReLU (0.2)

6 Computational Cost and Code Availability

All experiments were implemented in **PyTorch**. Each training run was performed on a single NVIDIA RTX A6000 GPU. To accelerate experimentation across short-scan durations, we executed multiple runs in parallel using up to eight GPUs.

Training for a single short-scan duration τ (including VAE pretraining, diffusion training, and reward-aligned fine-tuning) required approximately 40 hours on average. Because each duration is trained independently, the total wall-clock time can be significantly reduced through straightforward data parallelism across GPUs.

The code to run the SALD pipeline is available at <https://github.com/BenjamTorr/SALD>.

References

1. Clark, K., Vicol, P., Swersky, K., Fleet, D.: Directly fine-tuning diffusion models on differentiable rewards. In: Kim, B., Yue, Y., Chaudhuri, S., Fragkiadaki, K., Khan, M., Sun, Y. (eds.) International Conference on Learning Representations. vol. 2024, pp. 4793–4822 (2024), https://proceedings.iclr.cc/paper_files/paper/2024/file/13e8be77982beb73d7ed0bbf122f9f3c-Paper-Conference.pdf

Table 8: Training hyperparameters for the generative baselines.

Stage	Hyperparameter	Value
[CVAE]	Optimizer / LR	Adam / $3e-4$
[CVAE]	LR scheduler	ReduceLROnPlateau (0.5, patience 5)
[CVAE]	Batch size	32 (grad. accum. 2, effective 64)
[CVAE]	Loss	MSE + β -KL ($\beta = 2e-3$)
[CVAE]	Early stopping patience	15
[CVAE]	Random seed	1
[BicycleGAN]	Optimizer	Adam, betas (0.5, 0.999)
[BicycleGAN]	LR ($G+E$ / D)	$5e-5$ / $5e-5$
[BicycleGAN]	Batch size	32
[BicycleGAN]	Early stopping patience	25
[BicycleGAN]	Random seed	2
[Both]	Max epochs	200
[Both]	Gradient clipping max norm	1.0
[Both]	Mixed precision	bf16
[Both]	Samples per subject (inference)	10

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